

Tools Edition: Iris Investigate

INTRODUCTION

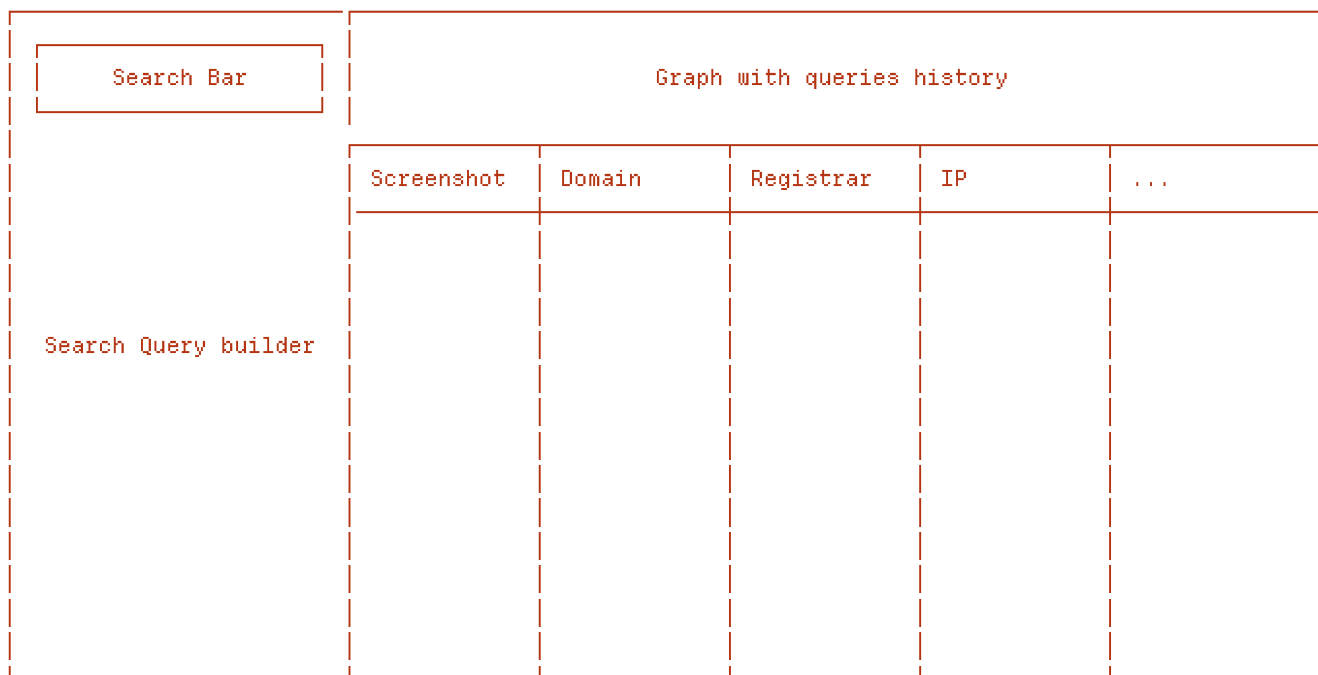
[Iris Investigate](#) is an investigation platform sold by DomainTools that is a goldmine when you want to investigate IP addresses and domain names. It aggregates data from various sources, including DNS, passive DNS, screenshots, web content, and domain registration data (WHOIS and RDAP), as well as TLS certificates. This data is presented in a way that allows you to identify pivot points.

I just want to clarify that I am simply an Iris Investigate user and am not affiliated with them in any way. I do, however, find their platform well designed, and I am trying to be transparent. It may sound like advertising, but it is not. Some valuable pivots are still hard to perform.

Also, this zine may contain inaccuracies due to the constrained format. Feel free to contact me if you have any questions or comments.

PIVOT ENGINE

The main view, or Pivot Engine, is organized as such:



- The search component usually does a pretty good job of identifying the data you paste in. The domain and IP are searched in the associated field, even if they are defanged, while free text is searched by Registrar by default. You can modify searches to add criteria with AND or OR logic. In addition to the graphical interface, Iris includes a SQL-like query language that can be used to easily share or modify queries in a textual format.
- To add criteria based on something you find interesting, right-click on any field or drag and drop it into the Query Builder. Iris will highlight values shared by only a few domains. This process is called "guided pivots" and, by default, shows values shared by 500 domains or fewer. This threshold can be changed for every field. Guided pivots are a good place to start when looking for a tracking heuristic!
- The graph view allows you to view and annotate previously made queries. This feature is useful because it allows you to track phishing

or malware families over extended periods while keeping the evolution of tracking queries accessible. I strongly suggest creating a dedicated investigation for each topic. Your future self will be grateful!

- The central widget display information on domains that match the query. It is recommended to go through the Widget setting to set the columns order. It is a matter of preferences but I like having the fields **First Seen, IP, Email Domain, Website Title** accessible as they are often pertinent pivoting field.

PIVOT ENGINE

Questions	Fields	Comments
Similar domains?	<code>IP,Website Title,Email,Registrar, etc.</code>	Search the list of domains and explore the pivot view to identify common patterns. Repeat until you find an effective query. If you suspect that false positives are still present, you can write a script or use other external sources, such as UrlScan, to validate the data.
DGA tracking?	<code>Domain,tld,Registrar, etc.</code>	Since DGA often relies on low-cost, lesser-used TLDs, you can usually perform wide queries based on TLDs and then refine the data (e.g., using regular expressions to match only the DGA domains). Some malware families also randomize registrar data in a way that allows tracking. (See also pDNS.)
Cloudflare tracking?	<code>Name Server</code>	It's common to come across domains protected by Cloudflare. Pairs of Cloudflare name servers associated with a registrar name or top-level domain (TLD) can sometimes be used to reliably identify clusters of domains. (See also Domain History)

DOMAIN AND WHOIS HISTORY

In addition to the Pivot Engine, Iris Investigate has other interesting tabs in an investigative context. For example, the Domain and Whois History tabs provide information about the evolution of a domain's hosting and ownership.

Questions	Fields	Comments
Domain reuse?	Created , Updated	In malvertising and SEO-related fraud, for example, attackers often reuse old domains shortly after their expiration date to leverage past traffic or trust. This view makes it easy to spot such reuse.
Cloudflare tracking?	IP	Sometimes domains are not protected with Cloudflare since their registration. As such, Domain History may contain domains real IP and thus provide a valuable pivot point.

PDNS

Iris aggregates several passive DNS (pDNS) sources. This allows powerful queries to track domain clusters related to specific subdomains or DGA, as some sources allow for regular expression searches. However, while RDAP/Whois is exhaustive regarding registered domains, pDNS depends on its sources. There are four sources for Iris, and each is described by a unique letter. These sources allow for different types of queries (supported DNS records, wildcard or regular expression support). They also have different telemetry/visibility, and as such, they can return distinct subdomains.

Questions	Infos	Comments
DGA tracking?	Scope, Query	If you are aware of uncommon words/patterns used under an uncommon TLD you can use regex to search domains <code><DGA regex>\.<TLD>\.\$</code> . Note that there is a mandatory trailing space at the end. Regex queries are resource-intensive, and sometimes it is easier to achieve similar results with wildcards.
Subdomains tracking?	Scope, Query	As for DGA, you can perform regex or wildcard queries. However, you will often have to filter by subdomain depth afterwards.
Subdomains enumeration?	Scope, Query	Common subdomains can be useful pivot points for confirming whether a domain is

Questions	Infos	Comments
		part of a cluster of domains. You can use regex query to search the subdomains of a few domains at a time. <pre>(domain1\ ..\ domainN)\.\$.</pre> For more domains, it is recommended to use external sources, such as The Hacker's Choice.